

CALCULATION OF MARGIN OF SAFETY FOR RAILROAD BONDS
(UNIT \$1,000; CALENDAR YEAR 1931)

Item	Chesapeake & Ohio	Chicago Great Western	Northern Pacific
1. Gross revenue	\$119,552	\$20,108	\$62,312
2. Net after taxes (railway operating income)	35,417	4,988	3,403
3. Equipment and joint-facility rents	dr. 88	dr. 2,417	cr. 3,398
4. Net after rents (net railway operating income)	35,329	2,571	6,801
5. Other income	2,269	196	16,853
6. Gross income	\$ 37,598	\$ 2,767	\$23,654
7. Interest and other fixed charges	10,902	1,866	14,752
8. Balance for dividends (net income)	\$ 26,696	\$ 901	\$ 8,902

Chesapeake and Ohio, 1931

Gross income exceeds net after taxes. Therefore use fixed-charges test (Test A).

$$\text{Fixed charges earned} = \frac{(6)}{(6) - (8)} = \frac{37,598}{10,902} = 3.45 \text{ times}$$

Chicago Great Western, 1931

Net after taxes exceeds gross income. Therefore use net-deductions test (Test B).

$$\text{Net deductions earned} = \frac{(2)}{(2) - (8)} = \frac{4,988}{4,087} = 1.22 \text{ times}$$

Northern Pacific, 1931

Gross income exceeds net after taxes. Therefore use fixed-charges test (Test A).

$$\text{Fixed charges earned} = \frac{23,654}{14,752} = 1.60 \text{ times}$$

NOTES ON THE FOREGOING TESTS

1. Chesapeake and Ohio represents the typical exhibit in which the results of both tests would have pointed to the same conclusion—in this case to the presence of a satisfactory margin of safety for the bonds.

2. In the case of Chicago Great Western, Test A, which is ordinarily applied, would not adequately reflect the burden of the unusually large rental deductions. Their effect is shown by Test B, and in accordance with our suggestion this less favorable result should be the one considered by the investor.

3. Northern Pacific presents the opposite situation. Its other income has been exceptionally large as compared with the bond interest, so that in most years the net deductions figure out as a credit. In this case the investor should follow the results of Test A, and consider Test B as a secondary indication of strength.